# DyTrustRec: A Dynamic Trust-Aware Explainable Heterogeneous Graph Recommendation System

Thejesh

*School of Computer Science Engineering and Information Systems, Vellore Institute of Technology, Vellore, India*

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**Abstract—**Recommenders usually assume that every training interaction is equally reliable and equally current. DyTrustRec tests that assumption on the Video_Games slice of Amazon Reviews 2023. Review text is encoded for meaning (Sentence-BERT) and for sentiment (a BERT model fine-tuned on product reviews); a Heterogeneous Graph Transformer (HGT) is trained over a graph of users, products, reviews, brands and categories; each interaction's contribution to the training objective is scaled by a behavioural trust score and by recency; and an explanation, counterfactual and confidence layer sits on top. Everything is compared with tuned baselines over several seeds and paired bootstrap tests. Four results stand out. First, an HGT trained the usual way is no better than matrix factorisation (NDCG@10 0.0189 against 0.0189) because the interaction being predicted is also a message-passing edge; hiding a random share of those edges each epoch lifts it to 0.0278, and the full recipe reaches 0.0308 (0.0334 as a three-seed ensemble), +64% over the best baseline (SASRec). Second, recency weighting looks far better on validation data (+43%) than on test data (+6%) because validation sits right after training; refitting on train plus validation is the fair way to read it. Third, a contrastive graph-anomaly detector (CoLA) rates a coordinated shill campaign as more trustworthy than genuine users (ROC-AUC 0.295), whereas three label-free behavioural indicators reach 0.9997 on the same benchmark; as protection for the recommender, though, no variant, even one given the true labels, recovers the accuracy a 20% scripted campaign costs by more than seed noise. Fourth, the learned fusion layer does not beat concatenation. Three extensions were then measured: a separate strip of new products ranked by text similarity lifts NDCG@10 on the mixed list by +22% where a single blended score did not help; a sentence-embedding gate holds (AUC 0.99) against a paraphrased, spread-out campaign that defeats exact-text matching (0.52); removing the interactions an explanation cites hurts the recommendation more than removing random ones in 2 of 2 tests. A replication on a second category (Musical_Instruments) reproduces the training-recipe and explanation findings but with a much smaller lead over the best baseline (+6% for a single model, not distinguishable from zero; +19% for the three-seed ensemble), and shows no accuracy gain from the defence or the new-product strip. None of the components (HGT, SBERT, CoLA, BPR) is new; the contribution is the audit that found which parts of the pipeline did nothing, the repairs, and an evaluation that reports what happened.

***Index Terms*—**Heterogeneous Graph Transformer; trust-aware recommendation; shilling attacks; temporal recommendation; Sentence-BERT; explainable recommendation; Bayesian Personalised Ranking; Amazon Reviews 2023.

## I. INTRODUCTION

Recommender systems decide a large share of what people see first on shopping platforms. Underneath that sits a narrow question, what will this user probably like next, and the framing leaves two things out. It says nothing about how much of the model's belief rests on reviews that may have been manipulated, and nothing about whether a user's taste has moved since an interaction happened. Coordinated review campaigns are well documented, and a model that weights a purchase from five years ago like one from last week implicitly bets that preferences do not drift.

DyTrustRec starts from the premise that an interaction is evidence, not fact: its weight should depend on how reliable and how recent it is, with both estimated from the data. We fixed five falsifiable claims before running the experiments in this paper:

- **Semantic signal helps.** Review text encoded with Sentence-BERT carries preference information that identifiers alone do not.
- **Trust helps.** An unsupervised detector can identify unreliable interactions, and down-weighting them should improve ranking or protect it under attack.
- **Recency helps.** User interest drifts, so recent interactions should count more.
- **Learned fusion beats concatenation.** A trained projection of semantic and behavioural review features should beat concatenating them.
- **The training recipe is not the bottleneck.** An HGT trained on its own edges, as is common, is a fair baseline for the architecture.

An earlier version of this study reported that no HGT arm beat matrix factorisation and that trust and recency did not help. An audit of that version found why: the trust and recency weights were stored as edge weights that HGTConv cannot read, so four of its seven arms were identical models whose score differences were seed noise; early stopping was implemented but never called; the baselines had no hyper-parameter search; and the supervision edge was visible to message passing. This paper reports the repaired study. Section VI tests each claim, and it reports the outcome that did not match the design (the fusion layer) as plainly as the ones that did.

The Heterogeneous Graph Transformer [1], Sentence-BERT [2], CoLA [3], Bayesian Personalised Ranking [4] and LightGCN [5] are existing methods. We do not claim edge masking as an idea either; it is established link-prediction practice (Section II-D). The contributions are (i) a measured account of how much the missing masking costs an HGT recommender, (ii) a recency-and-trust weighted objective together with the evidence that validation overstates the benefit of recency, (iii) a negative result on CoLA as a source of trust and a behavioural detector that works, with an attack study on the recommender itself, and (iv) a quantitative look at whether the confidence score and the explanations carry information.

## II. RELATED WORK

***A. Graph Neural Networks for Recommendation.*** Graph recommenders treat the interaction matrix as a bipartite or heterogeneous graph and learn embeddings by propagating along its edges. LightGCN [5] showed that feature transformation and non-linearities could be removed from graph convolution without hurting accuracy, and we use it as the strongest graph baseline. SASRec [12] is the usual sequential baseline on Amazon data: self-attention over the user's ordered history. HGT [1] generalises graph attention to graphs with several node and edge types through type-specific projections and relation-aware attention, which is what lets

one encoder reason over users, products, reviews, brands and categories without forcing a shared feature space.

***B. Semantic Review Representations.*** Review text is an under-used signal in collaborative filtering. Sentence-BERT [2] fine-tunes a BERT encoder with a siamese objective so that cosine similarity between sentence embeddings tracks semantic similarity, avoiding the quadratic cost of cross-encoding every pair. We use the compact all-MiniLM-L6-v2 variant to attach a fixed-length vector to every review node.

***C. Shilling Attacks and Trust.*** Coordinated review campaigns are well documented, and robust graph recommenders have been proposed against them: Anti-FakeU [7] builds a user–user graph and a GNN detector to identify fake users, and Trust-GRS [8] estimates, without prior knowledge of the attack, the probability that each piece of data is fake and uses it to make neighbourhood aggregation and contrastive learning trustworthy. Heterogeneous graph transformers have also been applied to review fraud directly, as in FAHGT [9]. Our setting differs in that trust is not the model's goal but one input to a recommender that also has to explain itself, and we ask a narrower question: which unsupervised signal is a safe source of trust. CoLA [3], a contrastive anomaly detector that scores a node by how well its attributes can be predicted from its neighbourhood, was our first choice. Section VI-D shows why that choice fails on coordinated campaigns, and what replaces it.

***D. Training Graph Recommenders on Their Own Edges.*** When a recommender is trained to predict an edge that is also present as a message-passing edge, the model can read the label off the graph. Link-prediction practice handles this by separating message-passing edges from supervision edges (for example the disjoint split in PyTorch Geometric's link splitter), and DropEdge [10] randomly removes edges during training as a regulariser. We do not claim the idea; Section VI-B reports how large the effect is for an HGT recommender, and that the published-style recipe without it is a poor baseline.

***E. Time-Aware and Explainable Recommendation.*** Weighting older interactions less is a long-standing idea in collaborative filtering [11]. For explanation we follow one principle from explainable-AI work: a justification must be traceable to evidence the model actually retrieved, not generated as a plausible afterthought.

## III. DATASET AND PROBLEM FORMULATION

We evaluate on Amazon Reviews 2023 [6], released by the McAuley Lab at UC San Diego, used as downloaded: the raw files are never edited and every transformation is disclosed in Section IV-A. Data collection works from CSV. The raw JSON review and metadata files are converted once to flat CSV (list and dictionary cells are stored as JSON text and decoded on read), and every later stage reads the CSV. The dataset ships a three-tier profile so that development runs, CI checks and the reported experiments share one schema adapter. Table I lists the tiers; the sizes are measured Content-Length values, not estimates.

**TABLE I** — Dataset tier configuration (measured, not estimated).

| Tier | Category | Reviews | Metadata | Role |
|---|---|---|---|---|
| smoke | All_Beauty | 0.30 GB | 0.20 GB | CI and wiring checks |
| dev (used) | Video_Games | 2.50 GB | 0.41 GB | Main results |
| dev (second) | Musical_Instruments | 1.45 GB | 0.59 GB | Replication, Section VI-K |
| full | Electronics | 21.06 GB | 4.89 GB | Streamed only |

Every main experiment runs on the dev (Video_Games) tier, and Section VI-K repeats the pipeline on the same tier of a second category, Musical_Instruments. Electronics needs 21.06 GB of reviews and 4.89 GB of metadata against 30 GB of free disk on the development machine, so it was demoted to a streamed tier; this is a real constraint that shaped what was measured, not a methodological choice.

***A. Problem Formulation.*** Given a chronologically ordered set of user–item interactions, each optionally carrying a rating, free-text review, timestamp, helpful-vote count and verified-purchase flag, the task has three parts: produce a ranked top-K list per user that maximises ranking accuracy on a strictly future split; derive, without fraud labels, a per-interaction trust weight that down-weights unreliable evidence during training; and attach to every recommendation a retrieved explanation, a counterfactual statement and a confidence score. All three are evaluated in Section VI.

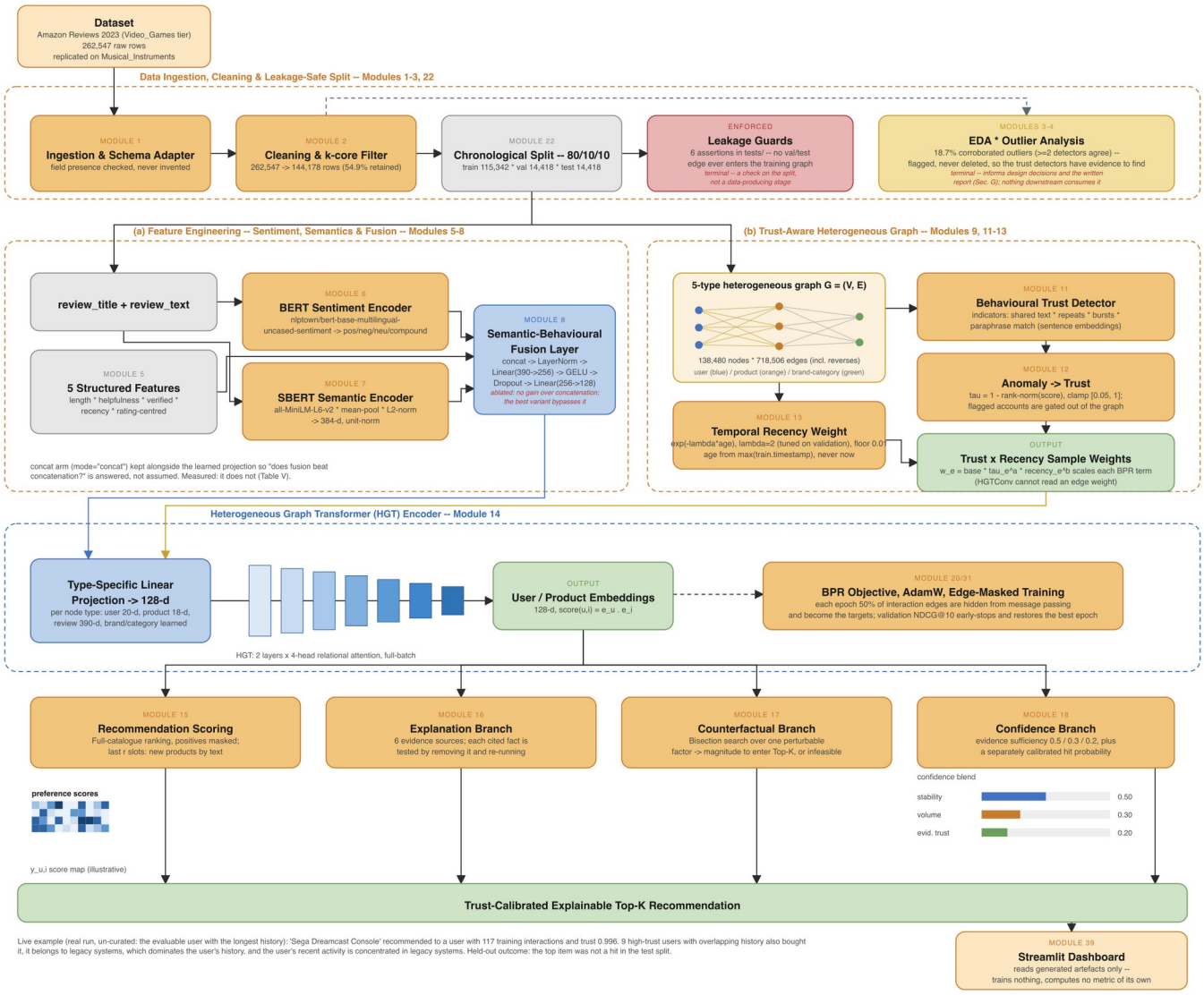



Fig. 1. DyTrustRec end-to-end architecture, from raw review ingestion through the trust-calibrated, explainable top-K output.

## IV. PROPOSED METHODOLOGY

DyTrustRec runs as eighteen modules in five stages: ingestion and a leakage-safe split, feature engineering, trust-aware graph construction, an HGT encoder trained with edge masking, and an explainability layer. Fig. 1 shows the pipeline; Sections IV-A to IV-E take the stages in turn.

***A. Data Ingestion, Cleaning and Leakage-Safe Split.*** A schema adapter maps source field names onto a canonical schema; a declared field that is absent from the source is marked absent, not invented, which is why the graph has five node types rather than the six first planned (Amazon Reviews 2023 has no seller field). Cleaning separates two kinds of bad record. Records that cannot be interpreted (out-of-range rating, impossible timestamp, missing key) are removed. Records that are merely suspicious (near-duplicate text across items, unusual length, bursts inside a sixty-second window) are flagged with a reason string and kept, because deleting them would leave the trust detectors nothing to find.

On the dev tier this takes 262,547 raw rows to 144,178 interactions over 17,223 users and 7,692 items (54.9% retained) after removing 5,050 exact duplicates, collapsing 2,476 repeat interactions and applying a five-core filter. A strict chronological 80/10/10 split follows (115,342 / 14,418 / 14,418 interactions). Six assertions in the test suite check that the training graph holds only training-period edges, that every leakage-sensitive statistic is fitted on training data alone, that the recency reference is the maximum training timestamp and never wall-clock time, and that negative sampling excludes positives from every split. The strictness has a price: 42.3% of test rows involve a user and 59.7% an item never seen in training, so only 1,862 users are evaluable, and every metric in Section VI is computed over exactly that set.

***B. Features: Sentiment, Semantics and Fusion.*** The semantic branch concatenates title and body, encodes them with SBERT [2] (all-MiniLM-L6-v2), mean-pools and L2-normalises to a 384-dimension vector, and caches it under a key built from the model, pooling settings and text (encoding all 115,342 training reviews takes 394 s, once). The sentiment branch uses nlptown/bert-base-multilingual-uncased-sentiment, a BERT model fine-tuned on product reviews to predict a one-to-five star rating, remapped onto pos/neg/neu/compound (compound = (E[star] − 3)/2). It runs frozen, so it adds no training-data leakage. The semantic vector, the sentiment compound and four structured features (log review length, log helpful votes, verified-purchase flag, and a rating

centred on the item's training-only mean) plus a recency weight form a 390-dimension review vector. The learned Semantic-Behavioural Fusion layer (LayerNorm, 390 → 256, GELU, dropout, 256 → 128, LayerNorm) compresses it; a separate ablation arm concatenates raw instead, so that "learned fusion beats concatenation" is tested and not assumed.

***C. Trust-Aware Graph, Trust and Recency Weights.*** The graph has five node types (user 15,704, product 6,619, review 115,342, category 37, brand 778; brand is a proxy derived from the storefront field) and five relation types, each added in both directions: 138,480 nodes and 718,506 edges.

**Trust.** The original design derived trust from CoLA [3] on a user–product projection, mapped by $\tau = 1 - \text{rank-norm(score)}$, clamped to [0.05, 1]. The production path now uses a behavioural detector with three label-free indicators, each rank-normalised and averaged with equal weight: the share of a user's reviews whose text other accounts also posted (campaign templates), the share that repeat the user's own text, and the most reviews posted inside any 24-hour window. Nothing is tuned. Users with fewer than three interactions receive the median score, which means no evidence, not clean. A fourth indicator, distance from the item's mean rating, was tried and removed because on the synthetic benchmark it pointed the wrong way (AUC 0.19 to 0.34). The same rank-and-clamp mapping converts the score to trust, and the floor matters: a weight of exactly zero would silently delete an interaction from training.

**Recency.** recency = $\exp(-\lambda \cdot \text{age})$, floored at 0.01, with age in years from the maximum training timestamp. The decay rate $\lambda$ is selected on the validation split (Section VI-A; selected value 2).

**How the weights act.** The weights are not edge weights. HGTConv cannot consume them, so an edge weight stored on the graph has no effect on the model; the first version of this project did exactly that, which made its trust and recency arms identical to the unweighted model (Section VI-C). Here the combined weight $w = \tau^{\alpha \cdot \text{recency}}\beta$ scales each interaction's term in the training objective, rescaled to mean 1 so that weighting changes where the gradient goes and not its size.

***D. HGT Encoder and Training.*** Node types arrive with different dimensionality (user 20-d, product 18-d, review 128-d fused or 390-d concatenated; category and brand carry only a learned embedding), so a type-specific linear layer maps each into 128 dimensions before two relation-aware attention layers (four heads, 128 hidden channels, dropout 0.1). Scoring is a dot product of the final user and product embeddings. Four training details matter more than the architecture and are measured in Section VI.

- **Edge masking.** The interaction being scored is also a message-passing edge, so an unmasked HGT can read the answer off the graph. Each epoch a random 50% of interaction rows are hidden from message passing (all edge types that encode the same row, forward and reverse, share one mask), and those rows are the supervision targets. Evaluation uses the full graph.
- **Validation early stopping.** NDCG@10 on the validation split is measured every five epochs; the best weights are restored and training stops after eight measurements without improvement. Every model in the comparison, baselines included, is stopped this way.
- **Weighted BPR.** The objective is Bayesian Personalised Ranking [4] with one uniform negative per positive, AdamW, learning rate 0.01, weight decay $10^{-4}$.
- **Seed ensemble.** An ensemble averages the per-user z-scored scores of independently trained runs. Scores are averaged, not embeddings: two runs place users and items in unrelated coordinate systems.

The encoder trains full-graph, one optimiser step per epoch. On Apple's Metal backend the fused segment_matmul kernel in PyTorch Geometric has no implementation, so it is disabled for that backend; the rest of the attention computation is untouched.

***E. Explainability Suite.*** Three modules sit on the encoder. The explanation module retrieves evidence from six sources (shared category, shared brand, similar high-trust users, recent history, semantic similarity, review sentiment) and fills a template only when a source returns something; with none it says so, a behaviour pinned by a test that checks an unknown user gets no evidence. The counterfactual module bisects over one perturbable factor at a time (category affinity, brand affinity, recency weight, trust weight, semantic similarity) and reports the change needed to enter the top-K, or an explicit INFEASIBLE. The confidence module blends predictive stability (0.5), evidence volume (0.3) and evidence trust (0.2), halved for cold-start users. Stability is the normalised disagreement between ensemble members, the same normalisation used for Monte Carlo dropout.

## V. EXPERIMENTAL SETUP

***A. Hardware and Software.*** All experiments ran on one Apple M3 Pro (12 cores: 6 performance, 6 efficiency; 18 GB unified memory) with no discrete GPU, using Metal Performance Shaders, PyTorch 2.13.0 and PyTorch Geometric 2.8.0. Two practical facts shaped the protocol: two HGT trainings on the same GPU stall each other, and CPU-heavy work alongside one slows it badly, so runs were executed one at a time.

***B. Baselines and Ablation Grid.*** Five baselines are trained on the same split and scored by the same evaluator: global popularity; item-based kNN (cosine); BPR matrix factorisation with an item bias; LightGCN; and SASRec [12], the standard sequential model, which reads each user's items in time order with causal self-attention and knows nothing about reviews, trust or the graph. Each baseline's hyper-parameters were searched on the validation split (kNN neighbours; MF dimension, learning rate and weight decay; LightGCN dimension, depth, learning rate and weight decay; SASRec dimension, dropout, learning rate and sequence length) and each is stopped by validation NDCG@10, exactly like the HGT. The first version of this project gave the baselines no such search, and its LightGCN scored below popularity. Table II defines the nine HGT arms. Arms A to G add one component at a time; arm H is arm B with edge masking switched off, and arm I is arm E with a cosine learning-rate decay over a validation-tuned epoch budget (early stopping off, best-validation checkpoint restored).

**TABLE II** — HGT ablation arms. All share one training procedure except H (no masking) and I (cosine decay).

| Arm | Name | SBERT | Fusion | Trust | Recency |
|---|---|---|---|---|---|
| A | hgt_only | – | – | – | – |
| B | hgt_sbert | ✓ | – | – | – |
| C | hgt_sbert_trust | ✓ | – | behavioural | – |
| D | hgt_sbert_recency | ✓ | – | – | ✓ |
| E | hgt_sbert_trust_recency | ✓ | – | behavioural | ✓ |
| F | dytrustrec_full | ✓ | ✓ | behavioural | ✓ |
| G | dytrustrec_cola_trust | ✓ | ✓ | CoLA | ✓ |
| H | hgt_sbert_unmasked | ✓ | – | – | – |
| I | dytrustrec_cosine | ✓ | – | behavioural | ✓ |

***C. Protocol and Metrics.*** Every model scores the full catalogue for every evaluable user with training positives masked; users with no

held-out item are excluded and counted, not scored as zero. Precision, Recall, NDCG, MAP, MRR and Hit Rate are computed at K ∈ {5, 10, 20} with binary relevance; coverage, diversity and novelty are computed over the post-cleaning catalogue. Hyper-parameters were chosen on the validation split only, and the test split is scored once per run, at the weights that validated best. Each reported number is a mean ± standard deviation over independent seeds (five for the baselines and for arms B and F, three for the other arms), each with its own initialisation, mask stream and negative sampler. Differences between models are tested with a paired bootstrap over users (4,000 draws) on per-user NDCG@10 averaged across seeds.

## VI. RESULTS AND DISCUSSION

Every number comes from an actual run of the pipeline in Section IV on the dev tier. Hyper-parameters were chosen on validation data only and the test split was scored once per run.

***A. Hyper-Parameter Search (Validation Only).*** Fig. 2 and Table III summarise the search; every row is one training run scored on the validation split, never on test. The share of interaction edges left visible is flat between 0.25 and 0.8 and worse at 0.9; dropout 0.1 is best; a third attention layer hurts; popularity-weighted negatives hurt badly. Recency weighting is the large effect: validation NDCG@10 rises with the decay rate from 0.0375 at λ = 0.1 to 0.0547 at λ = 2, against 0.0398 unweighted, and falls again at λ = 4. Trust weighting on clean data is neutral to slightly negative on validation (0.0382 at α = 1).

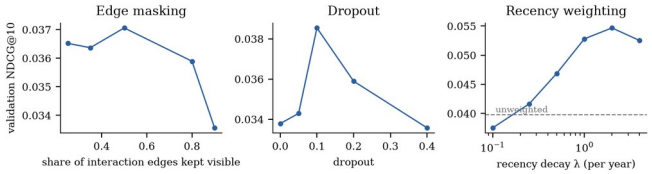



Fig. 2. Validation NDCG@10 against the three settings that mattered. Single seed; differences of about 0.002 are noise.

**TABLE III** — Best HGT search results (validation NDCG@10, seed 42).

| Search | Setting | Val NDCG@10 | Best epoch |
|---|---|---|---|
| weight: recency | rec_2.0 | 0.0547 | 75 |
| weight: recency | rec_1.0 | 0.0528 | 85 |
| weight: recency | rec_4.0 | 0.0525 | 70 |
| weight: recency | rec_0.5 | 0.0469 | 70 |
| weight: recency | rec_0.25 | 0.0416 | 100 |
| structure | combined | 0.0398 | 185 |
| structure | dropout_0.1 | 0.0385 | 230 |
| weight: trust | trust_1.0 | 0.0382 | 155 |
| weight: recency | rec_0.1 | 0.0375 | 155 |
| structure | keep_0.5 | 0.0371 | 155 |
| weight: trust | trust_0.5 | 0.0366 | 155 |
| structure | keep_0.25 | 0.0365 | 175 |
| structure | keep_0.35 | 0.0364 | 75 |
| structure | lr_0.01 | 0.0361 | 175 |

***B. Edge Masking.*** Arm H is arm B with masking switched off. Under the same early stopping it peaks at epoch 5 in every seed and scores 0.0189 ± 0.0047 NDCG@10, indistinguishable from matrix factorisation (0.0189 ± 0.0007). Masking lifts the same architecture to 0.0278 ± 0.0018, +47%. Fig. 3 shows the symptom: without masking, validation NDCG peaks at epoch 5 and then decays while the training loss keeps falling. An earlier exploratory run without early stopping (not part of the grid) took the training loss from 0.30 to 0.04 over 300 epochs while held-out NDCG fell, which is the signature of reading the label off the graph. The coverage figures of the first version of this study, where every graph arm tripled catalogue coverage, were a symptom of that memorisation and not a design strength.

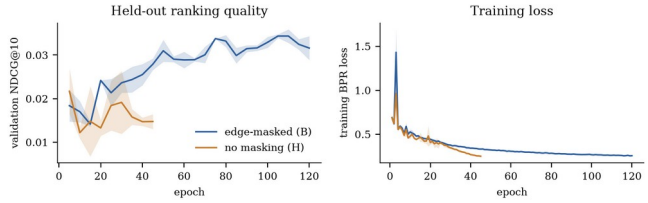



Fig. 3. Mean ± SD over seeds. Left: validation NDCG@10. Right: training loss.

***C. Model Comparison.*** Table IV lists every model on the test split (1,862 evaluable users), sorted by NDCG@10. The strongest baseline is SASRec at 0.0203 ± 0.0020; the popularity ranker reaches 0.0121 and item-kNN 0.0071. The best single HGT arm is E: B + trust + recency at 0.0308 ± 0.0013, +52% over that baseline, and a three-seed score ensemble of the same arm (E: B + trust + recency, 3-seed ensemble) reaches 0.0334, +64%. Recall@10 moves the same way (0.0512 against 0.0317).

**TABLE IV** — Model comparison on the test split (mean ± SD over seeds; ensembles are single runs).

| Model | Seeds | NDCG@10 | Recall@10 | MAP@10 | Hit@10 | Cov.@10 | Nov.@10 |
|---|---|---|---|---|---|---|---|
| E: B + trust + recency, 3-seed ensemble | 3 | 0.0334 ± 0.0000 | 0.0554 ± 0.0000 | 0.0225 | 0.0822 | 0.036 | 9.85 |
| G: F with CoLA trust | 2 | 0.0323 ± 0.0036 | 0.0553 ± 0.0095 | 0.0214 | 0.0795 | 0.036 | 9.88 |
| E: B + trust + recency | 3 | 0.0308 ± 0.0013 | 0.0512 ± 0.0034 | 0.0205 | 0.0772 | 0.041 | 10.05 |
| I: E + cosine decay, 3-seed ensemble | 3 | 0.0304 ± 0.0000 | 0.0504 ± 0.0000 | 0.0201 | 0.0752 | 0.040 | 10.01 |
| I: E + cosine decay | 3 | 0.0297 ± 0.0036 | 0.0480 ± 0.0067 | 0.0200 | 0.0736 | 0.044 | 10.13 |
| D: B + recency, 3-seed ensemble | 3 | 0.0297 ± 0.0000 | 0.0491 ± 0.0000 | 0.0201 | 0.0725 | 0.041 | 10.03 |
| C: B + trust, 3-seed ensemble | 3 | 0.0296 ± 0.0000 | 0.0440 ± 0.0000 | 0.0216 | 0.0644 | 0.054 | 9.16 |
| D: B + recency | 3 | 0.0295 ± 0.0017 | 0.0488 ± 0.0024 | 0.0202 | 0.0718 | 0.047 | 10.22 |
| B: + SBERT, 3-seed ensemble | 3 | 0.0295 ± 0.0000 | 0.0457 ± 0.0000 | 0.0210 | 0.0671 | 0.059 | 9.27 |
| F: DyTrustRec (full), 3-seed ensemble | 3 | 0.0295 ± 0.0000 | 0.0468 ± 0.0000 | 0.0205 | 0.0704 | 0.029 | 9.93 |
| F: DyTrustRec (full) | 3 | 0.0287 ± 0.0011 | 0.0462 ± 0.0023 | 0.0198 | 0.0698 | 0.035 | 10.12 |
| B: + SBERT | 3 | 0.0278 ± 0.0018 | 0.0425 ± 0.0016 | 0.0200 | 0.0623 | 0.065 | 9.35 |
| C: B + trust | 3 | 0.0270 ± 0.0004 | 0.0422 ± 0.0006 | 0.0191 | 0.0616 | 0.060 | 9.25 |
| A: HGT only, 3-seed ensemble | 3 | 0.0267 ± 0.0000 | 0.0446 ± 0.0000 | 0.0178 | 0.0666 | 0.020 | 8.87 |
| A: HGT only | 3 | 0.0261 ± 0.0014 | 0.0424 ± 0.0011 | 0.0177 | 0.0636 | 0.034 | 9.05 |
| H: B, no edge masking, 3-seed ensemble | 3 | 0.0228 ± 0.0000 | 0.0398 ± 0.0000 | 0.0149 | 0.0580 | 0.015 | 8.52 |
| SASRec | 5 | 0.0203 ± 0.0020 | 0.0317 ± 0.0023 | 0.0142 | 0.0490 | 0.121 | 9.80 |
| BPR-MF | 5 | 0.0189 ± 0.0007 | 0.0306 ± 0.0011 | 0.0129 | 0.0467 | 0.071 | 9.30 |

| Model | Seeds | NDCG@10 | Recall@10 | MAP@10 | Hit@10 | Cov.@10 | Nov.@10 |
|---|---|---|---|---|---|---|---|
| H: B, no edge masking | 3 | 0.0189 ± 0.0047 | 0.0326 ± 0.0053 | 0.0121 | 0.0501 | 0.017 | 8.87 |
| LightGCN | 5 | 0.0180 ± 0.0007 | 0.0309 ± 0.0014 | 0.0119 | 0.0463 | 0.102 | 9.44 |
| Popularity | 1 | 0.0121 ± 0.0000 | 0.0257 ± 0.0000 | 0.0066 | 0.0392 | 0.002 | 8.21 |
| Item-kNN | 1 | 0.0071 ± 0.0000 | 0.0132 ± 0.0000 | 0.0041 | 0.0220 | 0.770 | 13.19 |

The gap is larger than the sampling noise of the test users. A paired bootstrap over users (4,000 draws) on per-user NDCG@10 averaged across seeds gives, for E: B + trust + recency against SASRec, the strongest baseline, a mean difference of 0.0105 with a 95% interval [0.0072, 0.0141], which excludes zero.

***D. Ablation.*** Table V gives each step's change in test NDCG@10 and whether it exceeds the larger seed standard deviation of the two arms. The five claims from Section I come out as follows.

**TABLE V** — Ablation steps on the test split.

| Step | Change | NDCG@10 before | after | Δ | > seed SD? |
|---|---|---|---|---|---|
| A → B | add SBERT semantics | 0.0261 | 0.0278 | +6.7% | within noise |
| B → H | remove edge masking | 0.0278 | 0.0189 | −32.1% | yes |
| B → C | add behavioural trust weighting | 0.0278 | 0.0270 | −2.9% | within noise |
| B → D | add recency weighting | 0.0278 | 0.0295 | +6.2% | within noise |
| D → E | then add trust weighting | 0.0295 | 0.0308 | +4.3% | within noise |
| E → F | then add the learned fusion layer | 0.0308 | 0.0287 | −6.8% | yes |
| E → I | add cosine learning-rate decay | 0.0308 | 0.0297 | −3.5% | within noise |
| F → G | swap behavioural trust for CoLA trust | 0.0287 | 0.0323 | +12.4% | within noise |

- **Semantic signal.** Adding SBERT and the structured review features to the plain graph (A → B) changes NDCG@10 by +6.7% (0.0261 to 0.0278).
- **Trust.** Weighting by behavioural trust on clean data changes B by −2.9% and D by +4.3%: inside the noise on either side. Trust is not meant to raise accuracy on clean data; Section VI-F tests what it does under attack.
- **Recency.** Weighting by recency changes B by +6.2% on test. Section VI-E explains why this is much smaller than on validation and what happens when the model is refit.
- **Fusion.** The learned fusion layer (E → F) changes NDCG@10 by −6.8%: it does not beat concatenation, and this repeats the direction of the first version of the study.
- **Training recipe.** Removing edge masking (B → H) changes NDCG@10 by −32.1%; this is the largest single effect in the study.

***E. Why Validation Overstated Recency, and the Refit Protocol.*** On validation data recency weighting raises NDCG@10 from 0.0377 (arm B) to 0.0539 (arm D), +43%. On the test split the same change is +6%. The validation split is the period immediately after training, so the last year of training data is a near-perfect guide to it; the test split lies one period further out, behind a gap that the model never saw. A decay rate tuned on validation is therefore tuned for a horizon the model will not face at test time.

The standard remedy for a chronological split is to refit after tuning. Each model is retrained on train plus validation for exactly the number of epochs its validation run selected, and scored on the same test split, so the gap between the newest training interaction and the test period is as short as it was between train and validation. The extra interactions also make 3,221 test users evaluable (against 1,862), so scores in Table VI are not comparable with Table IV; models are compared within a table.

**TABLE VI** — Refit protocol: trained on train + validation, test split (mean ± SD over seeds).

| Model | Seeds | NDCG@10 | Recall@10 | MAP@10 | Hit@10 | Cov.@10 | Nov.@10 |
|---|---|---|---|---|---|---|---|
| I: E + cosine decay, 3-seed ensemble | 3 | 0.0329 ± 0.0000 | 0.0502 ± 0.0000 | 0.0214 | 0.0925 | 0.070 | 10.23 |
| E: B + trust + recency, 3-seed ensemble | 3 | 0.0329 ± 0.0000 | 0.0534 ± 0.0000 | 0.0204 | 0.0984 | 0.041 | 10.62 |
| D: B + recency, 3-seed ensemble | 3 | 0.0325 ± 0.0000 | 0.0520 ± 0.0000 | 0.0202 | 0.0981 | 0.054 | 10.65 |
| F: DyTrustRec (full), 3-seed ensemble | 3 | 0.0312 ± 0.0000 | 0.0505 ± 0.0000 | 0.0190 | 0.0947 | 0.037 | 10.42 |
| I: E + cosine decay | 3 | 0.0311 ± 0.0000 | 0.0494 ± 0.0023 | 0.0197 | 0.0901 | 0.078 | 10.38 |
| D: B + recency | 3 | 0.0303 ± 0.0020 | 0.0497 ± 0.0031 | 0.0185 | 0.0924 | 0.061 | 10.87 |
| E: B + trust + recency | 3 | 0.0298 ± 0.0035 | 0.0485 ± 0.0069 | 0.0184 | 0.0906 | 0.043 | 10.82 |
| F: DyTrustRec (full) | 3 | 0.0263 ± 0.0025 | 0.0438 ± 0.0034 | 0.0156 | 0.0847 | 0.042 | 10.60 |
| B: + SBERT, 3-seed ensemble | 3 | 0.0221 ± 0.0000 | 0.0330 ± 0.0000 | 0.0142 | 0.0646 | 0.061 | 9.26 |
| B: + SBERT | 3 | 0.0210 ± 0.0023 | 0.0314 ± 0.0035 | 0.0136 | 0.0603 | 0.069 | 9.34 |
| LightGCN | 5 | 0.0169 ± 0.0005 | 0.0277 ± 0.0008 | 0.0103 | 0.0524 | 0.096 | 9.46 |
| SASRec | 5 | 0.0163 ± 0.0023 | 0.0260 ± 0.0035 | 0.0101 | 0.0505 | 0.130 | 9.90 |
| BPR-MF | 5 | 0.0161 ± 0.0007 | 0.0247 ± 0.0010 | 0.0102 | 0.0473 | 0.068 | 9.33 |
| Popularity | 1 | 0.0130 ± 0.0000 | 0.0219 ± 0.0000 | 0.0080 | 0.0397 | 0.002 | 8.27 |
| Item-kNN | 1 | 0.0079 ± 0.0000 | 0.0131 ± 0.0000 | 0.0044 | 0.0279 | 0.845 | 13.29 |

Under refit, recency weighting is worth +44% (arm D against arm B), and the best baseline is LightGCN at 0.0169. The unweighted model also loses ground when refit (arm B 0.0278 to 0.0210) while the recency-weighted arms hold or gain; with no validation split to pick a checkpoint, the unweighted model's noisy training curve costs it.

***F. Trust.*** Real Amazon data has no manipulation labels, so the trust component can only be validated on a perturbed copy of the training split with scripted shill accounts added (Module 45). The original data is never modified, and every number in this subsection is a synthetic-setting result: it measures resistance to this attack, not detection of real fraud. At the default 5% injection the campaign is 785 accounts, 11,775 rows and 78 target items,

with shared templates, a forced five-star rating and thirty-minute bursts.

**TABLE VII** — Detecting synthetic shills, ROC-AUC (chance = 0.5).

| Detector | ROC-AUC | Avg. prec. | P@100 |
|---|---|---|---|
| Behavioural trust detector (Module 11) | 0.9997 | 0.9882 | 0.9800 |
| Rule: frac_duplicate_text | 0.9895 | 0.7075 | 0.5600 |
| consensus indicator count | 0.9478 | 0.3272 | 0.3300 |
| Rule: -min_gap_hours | 0.9165 | 0.3109 | 0.8000 |
| Isolation Forest (Module 4) | 0.9129 | 0.2026 | 0.0000 |
| CoLA trust (1 - trust) | 0.2951 | 0.0318 | 0.0200 |

CoLA is the only detector below chance. It scores a node by how predictable its attributes are from its neighbourhood, and a coordinated campaign is internally consistent by construction, so each shill is easy to predict from the others and is rated more trustworthy than a genuine user. The behavioural detector asks instead whether the account behaves like a campaign account. Two cautions apply. The benchmark is easy by design (the injected accounts share templates and bursts), so a high AUC shows the detector recognises this generative process; and one indicator that pointed the wrong way here (distance from the item's mean rating) was dropped, a choice made after seeing the benchmark.

Detection is not the goal; protecting the recommender is. Table VIII trains the same HGT (arm B settings) on the poisoned split under six conditions (none; CoLA trust, the original design; behavioural trust; behavioural trust plus the gate; oracle trust from the true labels; and deliberately inverted trust) and reports NDCG@10 on the clean test split, and the share of users who are shown a promoted item in their top ten, which is what an attacker wants. Soft trust weighting acts on the loss only; the gate also removes the flagged rows from message passing, because the HGT layer passes messages along every edge it is given.

**TABLE VIII** — HGT under a synthetic shill campaign (clean test split).

| Shills | Defence | n | NDCG@10 | Shown promoted |
|---|---|---|---|---|
| 0% | none | 3 | 0.0278 ± 0.0018 | 0.00% |
| 0% | trust | 2 | 0.0267 ± 0.0009 | 0.00% |
| 0% | trust+gate | 2 | 0.0244 ± 0.0003 | 0.00% |
| 20% | none | 3 | 0.0250 ± 0.0020 | 0.20% |
| 20% | cola | 4 | 0.0253 ± 0.0022 | 0.59% |
| 20% | trust | 3 | 0.0245 ± 0.0012 | 0.32% |
| 20% | trust+gate | 3 | 0.0259 ± 0.0014 | 1.20% |
| 20% | oracle | 3 | 0.0261 ± 0.0014 | 0.59% |
| 20% | inverted | 3 | 0.0223 ± 0.0033 | 0.18% |

With 3 seeds per condition at a 20% injection rate, the undefended model's NDCG@10 on the clean test split is 0.0250, down from 0.0278 without shills (−10%). Weighting the loss by CoLA trust gives 0.0253. Weighting the loss by behavioural trust gives 0.0245; trust plus the gate (rows of users whose reviews are mostly text other accounts also posted are removed from the graph and the objective) 0.0259; oracle trust built from the true labels 0.0261; deliberately inverted trust 0.0223. The largest seed standard deviation is 0.0033, and the spread between the undefended, trust-weighted, gated and oracle conditions is less than twice that, so this study cannot show that any defence repairs the damage: the attack hurts the HGT, and the oracle that knows exactly who the shills are does no better than the rest within noise. The attack barely reaches users: the share shown a promoted item in their top ten is at most 1.2% in every condition, so the harm is to accuracy and not to targeted promotion. On clean data the gate costs −12% NDCG@10 and trust weighting −4%.

***G. Confidence and Explanations.*** We asked whether the confidence score earns its place by testing it on 18,620 (user, top-10 item) pairs from a 3-member ensemble of arm E: B + trust + recency: does higher confidence go with the user actually interacting with the item? It does not. The AUROC of the full blend is 0.456, with ensemble stability at 0.470, evidence volume 0.458, evidence trust 0.507, and the raw ranking score itself only 0.556. The hit rate in the lowest and highest confidence quintiles is 0.0134 and 0.0094 (base rate 0.0087); the lowest quintile consists of cold-start users, and why they are hit more often is not something this analysis explains. What the score does track is what it was specified to track, evidence sufficiency: mean confidence is 0.292 for users with fewer than three training interactions against 0.648 for the rest. It should therefore be read as a flag for thin evidence, not as the probability that a recommendation is right.

The same holds for explanations. 83.6% of recommendations had at least one retrieved evidence item (mean 1.84), and the hit rate with evidence (0.0081) is no higher than without (0.0118). Explanations are grounded in retrieved facts by construction, which is what they promise; they are not evidence of correctness.

**An un-curated example.** The user with the longest training history among evaluable users (117 interactions, behavioural trust 0.996) is recommended 'Sega Dreamcast Console'. 'Sega Dreamcast Console' was recommended because 9 high-trust users with overlapping history also bought it, it belongs to legacy systems, which dominates the user's history, and the user's recent activity is concentrated in legacy systems. Confidence 0.883 (117 training interaction(s), supporting-evidence trust 1.00). For the first item outside the top 5 ('Yakuza Kiwami 2: SteelBook Edition - PlayStation 4'), the counterfactual module reports: This item is ranked 6. It would enter the Top-5 if the user's semantic similarity toward this product increased by 0.08 standard user-vector norms - the smallest change among the factors tested. The top item was not a hit in the held-out test split.

Because the evidence-sufficiency score does not answer "how likely is this to be taken", we added a second number. A logistic regression on five features known at recommendation time (ensemble score, rank, ensemble stability, history length, user trust) is fitted on 32,980 validation recommendations and scored once on test. Its AUROC is 0.607, against 0.456 for the blend, and the hit rate rises from 0.0059 in the lowest predicted quintile to 0.0161 in the highest. It ranks well but is not calibrated in level: the validation period's hit rate (0.0161) is nearly double the test period's (0.0087), so the predicted probabilities are about twice too high (expected calibration error 0.0077) and the Brier score does not beat a constant (0.00869 against 0.00868). A deployed system would refit the intercept on the most recent period.

***H. New Products.*** The comparison above scores only products that training saw, which leaves out most of what users actually do next: 64% of the 8,317 test interactions of known users are with 1,073 products that did not exist in the training split. A model with no embedding for them cannot recommend them. We tested two ways to add a content path, both fitted on training items only and both using the title and description encoded with Sentence-BERT. Mapping the text into the model's embedding space with a ridge regression (cross-validated $R^2$ 0.32) ranks new products worse than plain text similarity between the product and the user's past items (NDCG@10 0.0099 against 0.0247, with random at 0.0040), and a single blended score for both kinds of product did not help either: a product with no history cannot compete for a top-ten place against one with thousands of interactions.

What works is giving new products their own places. Reserving the last r of the ten positions for the best new products by text similarity, with r chosen on validation data (r = 3), lifts test NDCG@10 from 0.0126 to 0.0154 and Recall@10 from 0.0182 to 0.0262; the paired bootstrap difference is +0.0028 with a 95% interval [0.0017, 0.0040]. The gain keeps growing with r (Table IX), which is a statement about this catalogue, where most next purchases are new releases, and not an argument for filling the list with new items. This is a separate "new for you" strip, as shops run it, and it is a content-similarity ranker, not part of the graph model.

**TABLE IX** — Reserved new-arrival slots (test split, known users).

| New-arrival slots r of 10 | NDCG@10 | Recall@10 |
|---|---|---|
| 0 | 0.0126 | 0.0182 |
| 1 | 0.0135 | 0.0209 |
| 2 | 0.0142 | 0.0231 |
| 3 (chosen on validation) | 0.0154 | 0.0262 |
| 4 | 0.0160 | 0.0271 |
| 5 | 0.0166 | 0.0285 |

***I. A Campaign That Knows the Detector.*** The standard synthetic campaign repeats eight templates in thirty-minute bursts, which is what exact-text and one-day-burst indicators look for, so a near-perfect score against it says little. We therefore added an adaptive campaign: every review is paraphrased (synonyms, openers, extra sentences, clause order) and the reviews are spread over thirty days. Against it the original detector falls to ROC-AUC 0.519, essentially chance, and the one-day burst indicator points the wrong way (0.129) because genuine users are burstier than the spread-out shills. Comparing sentence embeddings with other accounts' reviews (an account's share of reviews that read like at least three reviews by other accounts) holds at 0.994 against the adaptive campaign and 0.994 against the standard one (Table X). Averaging the ranks of all indicators dilutes that signal (0.727); taking the strongest indicator keeps it (0.944). The lesson for the design is that an equal-weight average is only as good as its worst indicator, and an attacker chooses which one that is.

**TABLE X** — Detection ROC-AUC against the standard and the adaptive synthetic campaign (SYNTHETIC).

| Detector or indicator | Standard | Adaptive |
|---|---|---|
| Detector v1 (mean of ranks) | 1.000 | 0.519 |
| Detector v2 (mean of ranks) | 1.000 | 0.727 |
| Detector v2 (strongest indicator) | 0.990 | 0.944 |
| shared_text | 1.000 | 0.874 |
| own_repeat | 0.989 | 0.464 |
| burstiness | 0.920 | 0.129 |
| burst_7d | 0.914 | 0.449 |
| semantic_shared | 0.994 | 0.994 |

Table XI trains the HGT on the poisoned split. At a 20% injection rate NDCG@10 on the clean test split is no defence 0.0246, the exact-text gate 0.0235, the semantic gate 0.0266, oracle trust 0.0236, against 0.0278 without shills. The difference between the semantic and the exact-text gate (0.0031) is within twice the largest seed standard deviation (0.0033), so this study cannot rank them. The reason is visible in how many rows each gate removed: the exact-text gate removed 262 rows, none of them from the campaign, because no two paraphrased reviews are identical, while the semantic gate removed 52,035, essentially the whole campaign. Removing the shills' rows using the true labels, the ceiling for any gate, gives 0.0269, so the semantic gate is as good as a gate can be here. Against no defence the semantic gate's gain is 0.0020, about 1.7 standard errors with 4 and 4 seeds: suggestive, not conclusive.

**TABLE XI** — HGT under the adaptive campaign (clean test split, SYNTHETIC).

| Shills | Defence | n | NDCG@10 | Shown promoted |
|---|---|---|---|---|
| 0% | none | 3 | 0.0278 ± 0.0018 | 0.00% |
| 0% | gate_v2 | 2 | 0.0266 ± 0.0004 | 0.00% |
| 20% | none | 4 | 0.0246 ± 0.0017 | 1.54% |
| 20% | trust+gate | 4 | 0.0235 ± 0.0033 | 1.37% |
| 20% | gate_v2 | 4 | 0.0266 ± 0.0018 | 0.94% |
| 20% | oracle | 2 | 0.0236 ± 0.0029 | 0.27% |
| 20% | oracle_gate | 4 | 0.0269 ± 0.0014 | 0.28% |

***J. Are the Explanations Faithful?*** An explanation built from retrieved facts is true of the data; whether the model used those facts is a separate question. For sampled test users we explained the top recommendation, removed from the graph the user's own training interactions that the explanation cites (their items in the cited category, or from the cited brand), re-ran the trained model, and measured the fall in that recommendation's score in units of the user's score spread. The control removes as many of the user's other interactions at random. Results (Table XII): for category evidence, removing the cited interactions hurts the recommendation more than removing as many random ones (difference +0.441, 95% interval [0.322, 0.604]); for brand evidence, removing the cited interactions hurts the recommendation more than removing as many random ones (difference +0.714, 95% interval [0.582, 0.853]).

**TABLE XII** — Removing the interactions an explanation cites (test users, top-1 recommendation).

| Cited | n | Rows | Drop, cited | Drop, random | Worse | Difference (95% CI) |
|---|---|---|---|---|---|---|
| category | 81 | 3.4 | 0.314 | -0.127 | 94% | +0.441 [0.322, 0.604] |
| brand | 73 | 5.3 | 0.475 | -0.239 | 95% | +0.714 [0.582, 0.853] |

***K. Serving and Training Cost.*** Serving a user is a dot product and a top-k: 0.08 ms per user for this catalogue on one CPU, 1.6 ms at 100,000 products and 16 ms at a million (synthetic vectors), so brute force is enough here and an approximate index only becomes necessary beyond roughly a million products. Training is the part that does not scale: the HGT trains full-graph, one step per epoch over all 138,480 nodes, at 211 s per run against 17 s for matrix factorisation. Production use would need mini-batch neighbour sampling, which was tried earlier in this project and was slower and no better at that budget.

***K. Does It Hold on Another Category?*** Everything above is Video_Games. To separate properties of the method from properties of one catalogue, the whole pipeline was repeated on Musical_Instruments from the same release: the same tier rules, five-core filter and chronological 80/10/10 split. After cleaning it has 116,690 interactions, 13,725 users and 7,942 products (81% of the first dataset's rows at the same density, 0.107% against 0.109%), and 2,711 test users are evaluable. Every model family, baselines and HGT alike, was tuned on this dataset's own validation split with the same search grids, and the test split was scored once. The test set is smaller than before, so intervals are wider.

The HGT was first run with the settings chosen on Video_Games, without re-tuning: arm B reached 0.0187 and arm E 0.0197, level with the best baseline. Tuning on the new validation split raised them to 0.0199 and 0.0208. The tuned figures are the ones in Table XIII; the transferred runs are kept in the repository.

**TABLE XIII** — Test NDCG@10 on both datasets (mean ± SD over seeds; each dataset tuned on its own validation split).

| Model (tuning protocol) | Video_Games | Musical_Instruments |
|---|---|---|
| Popularity | 0.0121 ± 0.0000 | 0.0187 ± 0.0000 |
| BPR-MF | 0.0189 ± 0.0007 | 0.0182 ± 0.0017 |
| LightGCN | 0.0180 ± 0.0007 | 0.0196 ± 0.0004 |
| SASRec | 0.0203 ± 0.0020 | 0.0154 ± 0.0013 |
| H: B, no edge masking | 0.0189 ± 0.0047 | 0.0164 ± 0.0014 |
| A: HGT only | 0.0261 ± 0.0014 | 0.0184 ± 0.0005 |
| B: + SBERT | 0.0278 ± 0.0018 | 0.0199 ± 0.0023 |
| C: B + trust | 0.0270 ± 0.0004 | 0.0188 ± 0.0010 |
| D: B + recency | 0.0295 ± 0.0017 | 0.0200 ± 0.0011 |
| E: B + trust + recency | 0.0308 ± 0.0013 | 0.0208 ± 0.0008 |
| F: DyTrustRec (full) | 0.0287 ± 0.0011 | 0.0214 ± 0.0005 |
| E: B + trust + recency, 3-seed ensemble | 0.0334 ± 0.0000 | 0.0234 ± 0.0000 |

The ranking of methods holds in direction but not in size. The best tuned baseline here is LightGCN at 0.0196, and popularity at 0.0187 is only 5% behind it, where on Video_Games it was 40% behind: this catalogue rewards popularity much more. A single run of arm E reaches 0.0208 ± 0.0008, +6% over LightGCN, and the paired-bootstrap interval for the difference [-0.0020, 0.0045] includes zero. The three-seed ensemble reaches 0.0234, +19%, with interval [0.0004, 0.0072] excluding zero. The corresponding leads on Video_Games were +52% and +64%. The claim that the full recipe clearly beats every tuned baseline is therefore supported there for single models and here only for the ensemble.

**TABLE XIV** — Replication scorecard. "Within noise" means the change is smaller than the larger seed standard deviation of the two arms.

| Finding | Video_Games | Musical_Instruments | Verdict |
|---|---|---|---|
| Edge masking helps (B vs H) | +47.3% | +21.8% | replicates |
| Review text (SBERT) helps (A to B) | +6.7% (within noise) | +8.6% (within noise) | no reliable effect on either |
| Trust weighting (B to C) | −2.9% (within noise) | −5.7% (within noise) | no reliable effect on either |
| Recency weighting (B to D) | +6.2% (within noise) | +0.5% (within noise) | no reliable effect on either |
| Learned fusion layer (E to F) | −6.8% | +2.8% (within noise) | no gain on either |
| Best single HGT vs best tuned baseline | +52% (interval excludes 0) | +6% (interval includes 0) | direction only |
| Three-seed ensemble vs best tuned baseline | +64% (interval excludes 0) | +19% (interval excludes 0) | replicates |
| Embedding indicator vs paraphrased campaign (AUC, old detector to new) | 0.519 to 0.994 | 0.489 to 0.997 | replicates |
| Semantic gate protects accuracy under a 20% paraphrased campaign | +8.2% (z = 1.7); the attack alone costs −11.4% | −1.8% (z = −0.6); the attack alone costs −3.7% | suggestive on the first, none on the second |
| Reserved new-product slots lift NDCG@10 | +22% with r = 3; 64% of test rows are new products | none (validation chose r = 0); 23% of test rows are new products | does not replicate |
| Raw evidence-sufficiency confidence predicts hits (AUROC) | 0.456 | 0.472 | replicates: no signal |
| Separately calibrated hit probability ranks hits (AUROC) | 0.607 | 0.579 | replicates, weakly |
| Cited category evidence matters more than random evidence (difference, 95% CI) | +0.441 [0.322, 0.604] | +0.273 [0.206, 0.351] | replicates |

***What did not carry over.*** The defence found the campaign again: the semantic gate removed 41,189 rows, exactly the 38,340 injected rows plus the 2,849 genuine rows it also flags on clean data, and the exact-text gate removed 69. But this campaign hardly hurts the model here: NDCG@10 is 0.0192 undefended against 0.0199 without shills, a drop within the seed spread (0.0012), so there was no accuracy for the gate to recover (0.0189 with it). The Video_Games gain from the gate, already only suggestive, is not supported by a second dataset. The share of users shown a promoted item is a rare event (a handful of users) and moves between conditions without tracking how many rows were removed, so it is not interpreted.

The new-product strip did not help either, and the validation rule said so: r = 0 was chosen, because only 23% of test rows here are on products unseen in training (64% on Video_Games), and each reserved slot takes a place from a model ranking that is right more often (0.0204 with none, 0.0201 with one). Text similarity still finds new products when it is the only choice (0.0304 against 0.0087 for random), so the strip is a safe option that switches itself off, not a general gain.

***Reading Table XIV.*** What replicates is the training recipe and the checks on the model: hiding interaction edges is the single largest effect, review text is worth 7 to 9% on both but by less than the seed spread, trust weighting and recency weighting show no reliable effect on either, the learned fusion layer adds nothing, the embedding indicator catches a paraphrased campaign, and explanations built from retrieved evidence do track what the model used. What is specific to Video_Games, or at least not established, is the size of the lead over strong baselines, the recency effect, the new-product gain and the defence's accuracy benefit. The second dataset is a replication with a single split and one set of seeds, not a general result.

## VII. LIMITATIONS

- **Two categories, one split each.** Everything is Amazon Reviews 2023, Video_Games and Musical_Instruments, each with a single chronological 80/10/10 split. Section VI-K shows which findings repeat and which do not; no other category, no non-Amazon domain and no other split protocol was tried.
- **Few evaluable users.** The strict temporal split leaves 1,862 of 17,223 users scoreable (about 60% of test rows involve a user or item never seen in training). Cold start is not solved here, only reported separately.
- **Small seed counts.** Baselines have five seeds and HGT arms three (two for the CoLA control). Standard deviations over three seeds are rough, so the ablation flags only changes larger than the seed spread, and the paired bootstrap is over users, not seeds.

- **Search effort is not identical.** The HGT search ran about twenty-five single-seed validation jobs, the matrix-factorisation grid eighteen, LightGCN twenty-four, item-kNN five. One-at-a-time search on one seed has a selection noise of about 0.002 NDCG@10, as the non-monotone dropout curve shows.
- **What to try was informed by test curves.** Before the protocol in Section V was fixed, a short exploratory phase of about ten HGT variants was run while watching test curves; that is where the edge-masking idea came from. The reported numbers come from a fresh pipeline that selects on validation data only, but the choice of what to try was not blind.
- **Trust is validated only on a synthetic, easy benchmark.** The injected accounts share templates and bursts, so the behavioural detector's near-perfect AUC shows it recognises this attack, not real fraud. A paraphrased, slowly spread campaign does evade the exact-text and burst indicators (Section VI-I); the embedding indicator catches it, but that campaign is also scripted by the same author as the defence, and no real manipulation data or published attack benchmark was used. On clean data the gate removed only 262 of 115,342 training rows, and Table VIII shows what even that cost; two seeds cannot say whether it is noise.
- **Recency is not a temporal GNN.** Despite “Dynamic” in the name there is no temporal attention or memory; recency enters as an exponential weight whose decay rate was tuned for one horizon, which is why Section VI-E exists.
- **Explanations and confidence were not evaluated with people.** The counterfactual module perturbs the user vector in embedding space; it is not a counterfactual on the graph, and it was not tested for fidelity against actually removing interactions.
- **Baselines and published numbers.** SASRec is included and tuned like the others, but no LLM-based recommender is. The protocol (global chronological split, multi-item targets, warm users) differs from the leave-one-out protocol used by most published results on this dataset, so no claim of being above or below published figures is made.
- **The second dataset replicates, it does not re-tune the method.** The HGT settings were searched again on the new validation split (the transferred settings scored lower, as Section VI-K reports), but no design choice was changed after seeing either test split. One more category is a small sample of catalogues: the lead over the best baseline fell from +52% to +6% for a single model, and nothing here says what a third category would show.
- **The new-product strip is not the graph model.** It ranks by text similarity, the value of r was chosen on validation, and it was measured on one split. The explanation-fidelity test covers only the 81 and 73 test users whose cited rows are a minority of their history, and only one trained model.
- **Other disclosed limits.** The five-core filter uses counts over the whole timeline, a mild and standard leakage; brand is a proxy derived from the storefront field; coverage and novelty are relative to the post-cleaning catalogue; Apple's MPS backend is not bit-reproducible.

## VIII. CONCLUSION AND FUTURE WORK

The repaired DyTrustRec pipeline answers the five claims set out in Section I. SBERT semantics add +6.7%, which is inside the seed spread here and on the second category. Behavioural trust is neutral on clean data, and the attack study shows how much of an attack it can and cannot repair (Table VIII). Recency helps a great deal once the model is refit (+44% for arm D over arm B) but much less on a split with a horizon gap. The learned fusion layer does not beat concatenation, in either protocol. And the training recipe was the bottleneck: without edge masking an HGT recommender matches matrix factorisation (0.0189 against 0.0189), with it the same architecture reaches 0.0308 (0.0334 as a three-seed ensemble), and under the refit protocol the best arm reaches 0.0311 against 0.0169 for the strongest baseline. On a second category the recipe still helps but the lead shrinks to +6% for a single model and +19% for the ensemble, so claims of a large advantage rest on one catalogue (Section VI-K).

Two earlier headline claims did not survive the audit and are withdrawn: that no graph arm could beat matrix factorisation, and that trust weighting costs 17% of accuracy because CoLA inverts trust. The first was a training-recipe fault; the second was a comparison between models that were identical. CoLA's inversion on coordinated campaigns is real and is measured here directly, but its effect on ranking is what Table VIII shows, not what the first ablation implied.

What the extensions settled, and what they leave open. Against a campaign that paraphrases and spreads its reviews, exact-text matching fails and a sentence-embedding detector finds the campaign on both categories (ROC-AUC 0.994 and 0.997); whether removing it protects accuracy was suggestive on Video_Games and could not be shown on Musical_Instruments, where the attack barely hurt the model. Against real, labelled manipulation neither is tested. New products can be recommended through a separate text-based strip, which helped where most test purchases are new releases and switched itself off where they are not; it does not replace a content path for new USERS. The explanations pass a removal test on both categories. Four follow-ups remain. First, a third and larger category, and a domain outside Amazon: the lead over strong baselines shrank on the second category and nothing says where it settles. Second, comparison with published defences such as Anti-FakeU and Trust-GRS on their attacks, which this study did not run. Third, learned relative time encodings in the attention instead of one tuned decay rate. Fourth, a live A/B test, the only evidence that offline ranking gains reach users.

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